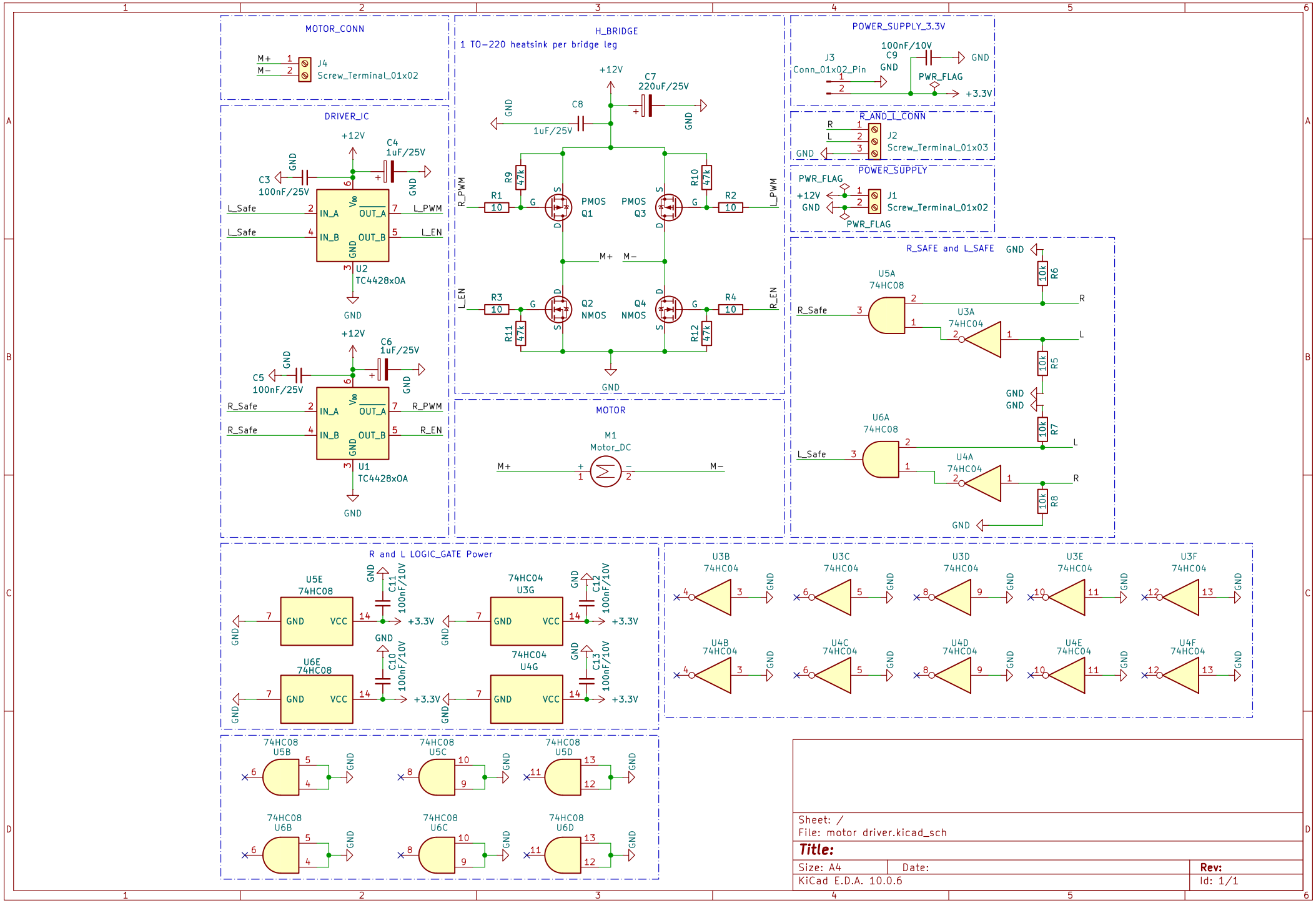

MOTOR_CONN
M+ 1
M− 2
J4
Screw_Terminal_01x02
DRIVER_IC
+12V
C4
1uF/25V
C3
100nF/25V
GND
L_Safe
L_Safe
IN_A
IN_B
OUT_A
OUT_B
L_PWM
L_EN
U2
TC4428xOA
R_Safe
R_Safe
R_PWM
R_EN
C6
1uF/25V
C5
100nF/25V
U1
TC4428xOA
H_BRIDGE
1 TO−220 heatsink per bridge leg
+12V
C7
220uF/25V
C8
1uF/25V
R9
47k
R10
47k
R1
10
R2
10
PMOS
Q1
PMOS
Q3
M+
M−
R3
10
R4
10
R11
47k
R12
47k
Q2
NMOS
Q4
NMOS
MOTOR
M1
Motor_DC
POWER_SUPPLY_3.3V
J3
Conn_01x02_Pin
100nF/10V
C9
PWR_FLAG
+3.3V
R_AND_L_CONN
R
L
J2
Screw_Terminal_01x03
POWER_SUPPLY
PWR_FLAG
+12V
J1
Screw_Terminal_01x02
R_SAFE and L_SAFE
U5A
74HC08
U3A
74HC04
R6
10k
R5
10k
R7
10k
R8
10k
U6A
74HC08
U4A
74HC04
R and L LOGIC_GATE Power
U5E
74HC08
U6E
74HC08
74HC04
U3G
74HC04
U4G
C11
100nF/10V
C12
100nF/10V
C10
100nF/10V
C13
100nF/10V
GND
VCC
+3.3V
U3B
74HC04
U3C
74HC04
U3D
74HC04
U3E
74HC04
U3F
74HC04
U4B
74HC04
U4C
74HC04
U4D
74HC04
U4E
74HC04
U4F
74HC04
74HC08
U5B
U5C
U5D
U6B
U6C
U6D
Sheet: /
File: motor driver.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 1/1